



AI 2002 – Artificial Intelligence Assignment 1

Group Members

Ayesha Naeem	21F-9190
Ayesha Zahid	21F-9136

Question No 01:

Examine the AI literature to discover whether the following tasks can currently be solved by computers. If yes, then list the name of a successful application or otherwise, state a reason why a specific task is not possible yet.

i. Drive safely on a highway.

Yes, computers can currently drive safely on highways.

Successful applications:

Tesla and Waymo.



Figure 1.1



Figure 1.2

ii. Drive safely in a busy city with other traffic and pedestrians.

Yes, computers can currently drive safely in busy cities with other traffic and pedestrians.

Successful applications:

1. Waymo
2. Tesla
3. Uber
4. Apple (Project Titan)

iii. Buying a week's worth of groceries online.

Yes, buying a week's worth of groceries online can currently be done by computers.

Successful applications:

1. Amazon Fresh
2. Instacart, etc

iv. Buying a week's worth of groceries at a superstore.

Yes, buying a week's worth of groceries at a superstore can be facilitated by computers.

Successful applications:

1. Walmart
2. Target, etc

v. Booking a doctor's appointment online.

Yes, booking a doctor's appointment online is a task that can currently be accomplished through various online platforms and healthcare systems.

Successful applications:

1. Zocdoc
2. Healthgrades
3. MyChart
4. Doctolib
5. Practo

vi. Win a soccer game against human(s).

No, computers and AI are not able to win a soccer game against human players.

Reason:

The main reason AI hasn't won soccer games against humans is because the real-world soccer is too complicated for AI. The game involves physical moves, changes in the environment, and the need to adapt quickly. AI is not good at these things yet. It lacks the physical abilities, quick adjustments, and complex coordination required to beat humans in a real soccer game.

vii. Win an art competition.

AI has not won any art competitions.

Reason:

AI struggles with replicating the depth of human creativity, emotional expression, and cultural nuances crucial for success in subjective domains like art competitions.

viii. Win a chess game against human(s).

Yes, AI has won a chess against humans.

Successful applications:

1. Stockfish
2. AlphaZero

ix. Discovering and proving new mathematical theorems.

No, Discovering and proving new mathematical theorems is a task that is currently not solved by computers.

Reason:

While AI can assist in mathematical research and problem-solving, discovering and proving new mathematical theorems typically relies on human insight, creativity, and intuition, which current AI lacks.

The reasoning and abstract thinking involved in mathematical discovery often go beyond the capabilities of existing AI systems.

x. Giving competent legal advice in a specialized area of law.

No, AI systems can only assist in legal research and provide information.

Reason:

AI lacks the experience, ethical judgment, and comprehensive understanding needed to provide competent legal advice in highly specialized areas of law. Giving competent legal advice in a specialized area of law requires a deep understanding of complex legal principles, ethical considerations, and contextual variation.

xi. Translating spoken Urdu into spoken English in real time.

Yes, AI can translate spoken Urdu into spoken English in real time.

Successful applications:

1. Google Translate
2. Microsoft Translator

xii. Writing an intentionally funny story.

Yes. AI can generate intentionally funny stories by processing and understanding patterns in humor from vast amounts of training data. It can produce narratives with jokes, wordplay, and comedic elements.

Successful applications:

1. ChatGPT
2. Jukebox.

xiii. Writing a publishable research article.

No. AI cannot write a publishable research article.

Reason:

AI cannot create a truly publishable and scientifically rigorous research article because it involves original research, domain expertise, critical analysis, and validation by human researchers. AI lacks the capacity for independent research and the variation understanding necessary for academic contributions.

Question No 2:

For each of the following assertions, say whether it is true or false and support your answer with examples or counterexamples where appropriate.

- i. An agent that senses only partial information about the state cannot be perfectly rational.**

False.

Perfect rationality refers to the ability to make good decisions given the sensor information received.

- ii. There exist task environments in which no pure reflex agent can behave rationally.**

True.

A pure reflex agent ignores previous percepts, so cannot obtain an optimal state estimate in a partially observable environment.

- iii. There exists a task environment in which every agent is rational.**

True.

For example, in an environment with a single state, such that all actions have the same output, it doesn't matter which action is taken.

- iv. Every agent function is implementable by some program/machine combination.**

False.

An agent function is an abstract mathematical description while the agent program is a concrete implementation running within some physical system. Since the agent function is just an abstract description it is completely possible that there exist cases in which an agent program will fail due to it running out of memory.

- v. **Suppose an agent selects its action uniformly at random from the set of possible actions. There exists a deterministic task environment in which this agent is rational.**

True.

This is a special case of an environment with a single state, such that all actions have the same output, if it doesn't matter which action you take, selecting randomly is rational.

- vi. **It is possible for a given agent to be perfectly rational in two distinct task environments.**

True.

For example, we can arbitrarily modify the parts of the environment that are unreachable by any optimal policy as long as they stay unreachable.

- vii. **Every agent is rational in an unobservable environment.**

False.

For example, the vacuum agent that cleans. If the agent moved but did not clean, it would not be rational.

- viii. **A perfectly rational poker-playing agent never loses.**

False.

A perfectly rational poker-playing agent can make optimal decisions based on available information, but it does not guarantee immunity to losses due to the inherent luck and hidden information in the game of poker.

Question No 3:

For each of the following activities, give a PEAS description of the task environment in tubular form.

Playing soccer

P	E	A	S
Teamwork Efficiency	Soccer field	Player's legs	Player Eyes
Tackles Made	The ball	Arms	Ears
Goals Scored	Opposing team	Head	Crowd Noise
Passes Completed	Teammates	Referee whistles	
Ball possession	Referee		
strategy	Boundary Lines		

Exploring the subsurface oceans of Titan

P	E	A	S
Safety	Ocean	Mobile Driver	Vedio Camera
Image quality	Water	Steering	Accelerometer
Vedio quality	Hazards/obstacles	Brakes	Depth Sensor
		Accelerator	GPS
		Engines	Sonar, radar

Playing a tennis match

P	E	A	S
Win the match	Tennis court	Legs	Eyes
Skill development	Ball	Arms	Ears
Sportsmanship	Opponent	Racket	Opponent Locator
Strategy	Scoring system		Scoreboard
	Crowd		Ball Locator

Practicing tennis against a wall

P	E	A	S
Tennis skills	Tennis court	Legs	Eyes
Consistency	Wall	Arms	Ears
Accuracy	Ball	Racket	Touch and feel
Power	Racket	Feet	

Performing a high jump

P	E	A	S
Safety	Wall	Legs	Eyes
Altitude	wind	Arms	Ears
	High jump bar	core	Camera
	landing mat	takeoff foot	Height Sensor
	temperature	jumping foot	
	runway		

Knitting a sweater

P	E	A	S
Quality	Knitting needles	Hands	Eyes
Completion Time	yarn	fingers	pattern instructions
Comfort	pattern	needles	
	lighting		
	distractions		

Bidding on an item at an auction

P	E	A	S
Cost	Auction house	Voice	Eyes
Value	auctioneer	bidding paddle	ears
Necessity	item	online interface	online information
Quality	bidding rules		
	opening bid		

Question No 4:

For each of the activities listed in the previous question, describe the environment type as:

- i. Fully or partially observable
- ii. Deterministic or stochastic
- iii. Episodic or sequential
- iv. Static or dynamic
- v. Discrete or continuous
- vi. Single or multi-agent

Playing Soccer

Fully or partially observable	The soccer field is partially observable as players can only see a limited portion of the field and must rely on memory, team communication, and other cues to infer the state of the rest of the environment
Deterministic or stochastic	The soccer game is stochastic due to unpredictable factors that can affect the outcome, such as the ball's movement, other players' actions, and weather conditions.
Episodic or sequential	The soccer game is sequential as each player's actions can affect the state of the environment and the actions of other players, creating a complex chain of cause-and-effect throughout the game.
Static or dynamic	The soccer game is dynamic , requiring players to adapt their strategies and tactics in real time due to the rapidly changing and unpredictable nature of the game.
Discrete or continuous	The soccer game is discrete , divided into distinct units of space (the field, the goals, the ball) and time
Single or multi-agent	The soccer game involves two teams of players, each with their own goals, strategies, and interactions, making it a multi-agent environment .

Exploring the subsurface oceans of Titan

Fully or partially observable	The exploration of Titan's subsurface oceans is partially observable since exploration vehicles can only gather limited data and must rely on memory, communication, and other instruments to understand the broader environment.
Deterministic or stochastic	The exploration environment is stochastic due to unpredictable factors such as the composition and behavior of the ocean, the presence of hazards, and the reliability of the exploration vehicle and its instruments
Episodic or sequential	The exploration is sequential as the vehicle's actions can impact the environment, creating complex cause-and-effect relationships over the course of the mission
Static or dynamic	The environment is dynamic , requiring the vehicle to adapt its strategy in response to the ever-changing ocean and the data it collects.
Discrete or continuous	The exploration takes place in a continuous flow of time and space, requiring real-time data collection and the ability to respond to environmental changes as they occur.
Single or multi-agent	The exploration mission could involve a single vehicle or multiple vehicles working together, depending on the specific exploration objectives and mission setup.

Playing a tennis match

Fully or partially observable	Partially observable: Tennis players can only see a limited part of the court and must rely on memory, communication, and other cues to infer the state of the rest of the environment.
Deterministic or stochastic	Stochastic: Factors such as the ball's movement, opponent's actions, and weather conditions make the tennis match environment stochastic, affecting the match's outcome.
Episodic or sequential	Sequential: Each player's actions can influence the state of the environment and the match's

	progression, creating complex cause-and-effect relationships.
Static or dynamic	Dynamic: The rapidly changing and unpredictable nature of the game requires players to adapt their strategies and tactics in real time.
Discrete or continuous	Discrete: The tennis match environment is divided into distinct units of space and time, such as the court, the ball, and the duration of the match
Single or multi-agent	Multi-agent: Involving two players (or four players in doubles), each with their own goals and strategies, makes tennis a multi-agent environment

Practicing tennis against a wall

Fully or partially observable	Partially observable: Practicing tennis against a wall is partially observable as the player can only see a limited portion of the wall, relying on memory and cues.
Deterministic or stochastic	Deterministic: The behavior of the wall is predictable, not influenced by the player's action
Episodic or sequential	Sequential: The player's actions impact the outcome, creating a cause-and-effect chain during the practice session
Static or dynamic	Static: The environment does not change over time except for the ball's movement and position
Discrete or continuous	Discrete: Divided into distinct units of space and time
Single or multi-agent	Single-agent: Primarily involves a single player, although multiple players can cooperate.

Performing a high jump

Fully or partially observable	Partially Observable: The high jump environment is partially observable as the jumper can only see a limited portion of the area and must rely on memory and cues to infer the state of the rest of the environment.
--------------------------------------	---

Deterministic or stochastic	Deterministic: The high jump environment is deterministic as the behavior of the jump area and the bar is predictable, unaffected by the jumper's actions.
Episodic or sequential	Episodic: Each jump is a self-contained episode with a clear beginning and end
Static or dynamic	Static: The high jump environment is generally static, with minimal changes over time.
Discrete or continuous	Discrete: The environment for performing a high jump is divided into distinct units of space and time.
Single or multi-agent	Single-agent: The high jump is generally a single-agent activity, involving only the jumper and potentially other observers or officials

Knitting a sweater

Fully or partially observable	Partially observable: Knitter sees limited sweater portion, relies on memory.
Deterministic or stochastic	Deterministic: Yarn and needles' behavior is predictable.
Episodic or sequential	Sequential: Knitter's actions affect environment, outcome.
Static or dynamic	Static: Yarn and needles remain unchanged.
Discrete or continuous	Discrete: Process divided into space/time units.
Single or multi-agent	Single-agent: Typically involves one knitter

Bidding on an item at an auction

Fully or partially observable	Partially observable: Bidder observes current auction state, infers past/future
--------------------------------------	--

Deterministic or stochastic	Stochastic: Outcome depends on bidders, item value, rules.
Episodic or sequential	Sequential: Bidding is discrete actions influencing auction.
Static or dynamic	Dynamic: Auction state changes rapidly, unpredictably.
Discrete or continuous	Discrete: Time/space divided into units
Single or multi-agent	Multi-agent: Involves multiple bidders with goals

Question No 5:

Pick a real-world example of your choice (a robot or software agent/bot, except for the examples discussed in detail in the class e.g., taxi & vacuum-cleaner agents) and describe different scenarios in the context of the following agent types that solve the problem.

- i. Reflex agent
- ii. Model-based.
- iii. Goal-based
- iv. Utility-based
- v. Learning agent

Real-World Example: Robotic Package Delivery System - Amazon Scout Delivery Robot

Reflex Agent

Scenario:

- When encountering an obstacle, such as a parked car, the reflex agent immediately triggers the robot to stop and change direction to avoid the obstacle.
- If the robot detects a pedestrian in front of it, the reflex agent prompts an instant stop to ensure the safety of the pedestrian and the robot.

Model-Based Agent

Scenario:

- The model-based agent uses a detailed map of the delivery area to plan an efficient route for the Amazon Scout Delivery Robot, taking into account obstacles, traffic patterns, and road conditions.
- If a new construction zone appears in the map's vicinity, the model-based agent dynamically updates the delivery route to circumvent the construction zone and optimize delivery time.

Goal-Based Agent

Scenario:

- The goal-based agent aims to minimize delivery time and maximize customer satisfaction by prioritizing nearby deliveries and optimizing package allocation to efficiently use the robot's capacity.
- If the robot encounters a situation where it can choose between delivering a package to a customer located nearby or one further away, the goal-based agent guides it to prioritize the closer customer for faster delivery.

Utility-Based Agent

Scenario:

- The utility-based agent assesses different actions based on a utility function that includes delivery time, energy consumption, and customer satisfaction.
- When the robot faces a detour due to construction, the utility-based agent evaluates the trade-offs between potential delays and the risk of damage to the robot or customer property, making decisions to optimize overall utility.

Learning Agent

Scenario:

- The learning agent continuously learns from its interactions and feedback to improve its delivery performance over time.
- As the Amazon Scout Delivery Robot completes deliveries in a new neighborhood, it uses machine learning algorithms to adapt its navigation and delivery strategies, becoming more efficient and effective with each delivery.

Question No 6:

A problem can be described as a “search problem” by specifying the following components: (a) initial state, (b) actions, (c) transition model/successor function, (d) goal state, (e) path cost (assume a path cost of 1 per action, if not specified explicitly). For each of the following problems, identify the above components.

- i. There are six glass boxes in a row, each with a lock. Each of the first five boxes holds a key unlocking the next box in line, the last holds a banana. You have the key to the first box, and you want the banana.**

(a) Initial State:

When we are the first glass box with the key of first box lock.

(b) Actions:

Opening the lock of first box is action.

(c) Transition Model/Successor Function:

Given a box with a key, unlock it to get the key for the next box.

(d) Goal State:

The goal state is the last box having the banana.

(e) Path Cost:

Assume path cost of 1 per action.

- ii. **Consider a 3-coins problem, C1-C2-C3 where each coin has two sides: a head (or H) and a tail (or T). Only one coin can be flipped at a time which will change an H to T or a T to H. Since, we have three coins, the coin position would tell the coin to be flipped, that is, C1 will flip coin 1 and so on. The initial state is H-H-H and the goal is to get at least two tails.**

(a) Initial State:

H-H-H

(b) Actions:

Flipping a coin.

(c) Transition Model/Successor Function:

The successor function generates the next state by flipping one of the coins.

(d) Goal State:

The goal state is when at least two coins have tail side.

(e) Path Cost:

Assume path cost of 1 per action.

iii. In the “missionaries and cannibals” problem, three missionaries and three cannibals must cross a river using a boat which can carry at most two people, under the constraint that, for both banks, if there are missionaries present on the bank, they cannot be outnumbered by cannibals (if they were, the cannibals would eat the missionaries). The boat cannot cross the river by itself with no people on board.

(a) Initial State:

All three missionaries and three cannibals are on one side of the river, and the boat is on the same side.

(b) Actions:

Moving two people either two missionaries, two cannibals, or one of each to the other side of river using the boat.

(c) Transition Model/Successor Function:

The successor function generates the next state by moving two people from one side to the other.

(d) Goal State:

The goal state is when all three missionaries and three cannibals are on the opposite side of the river, with the boat. The arrangement on both sides must also satisfy the constraint that missionaries cannot be outnumbered by cannibals on either side.

(e) Path Cost:

Assume a path cost of 1 per action.

Question No 7

For each of the problem in the previous question:

- i. Construct a state space and clearly show the initial state(s), possible actions, and goal state(s).**

i. Glass Boxes Problem:

State Space:

Initial State:

Locked1-locked2-locked3-locked4-locked5-Banana

Actions:

Unlock1, Unlock2, Unlock3, Unlock4, Unlock5

Goal State:

Unlocked1-Unlocked2-Unlocked3-Unlocked4-Unlocked5-Banana

ii. 3-coins Problem:

State Space:

Initial State:

H-H-H

Actions:

FlipC1, FlipC2, FlipC3

Goal State:

TT-H or TTT or THT or HTT

iii. Missionaries and Cannibals Problem:

State Space:

Initial State:

3Missionaries

3Cannibals

boat

Actions:

Move 2Missionaries,

Move 2Cannibals,

Move 1Missionaries -1Cannibals,

Move 1Missionaries, Move 1 Cannibal.

Goal State:

On the other side of river → 3 Missionaries, 3Cannibals and Boat.

ii. State Space Type:

The state space for the glass boxes problem is a tree.

The state space for the 3-coins problem is a tree.

The state space for the missionaries and cannibals' problem is a graph.